Heart Disease Predictor Using QNN

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***Abstract*—Heart disease remains one of the leading causes of death worldwide, with early detection being crucial for effective treatment and prevention. This study presents a Quantum Neural Network (QNN)-based approach for predicting the likelihood of heart disease using patient medical data. The proposed hybrid model combines classical preprocessing with a quantum variational circuit layer to exploit the representational advantages of quantum computation. The model processes input features such as age, sex, blood pressure, cholesterol levels, and other key clinical indicators to generate a probability score indicating the risk of heart disease. By leveraging quantum-enhanced transformations and non-linear feature mappings, the QNN effectively captures complex correlations in the data that traditional machine learning methods might overlook. Evaluation results demonstrate that the proposed model achieves an accuracy of 82.88%, an F1-score of 0.848, and an AUC score of 0.88, indicating balanced classification performance. This approach has the potential to contribute to early diagnosis, enabling timely medical interventions and reducing mortality rates.**

Keywords—Quantum Neural Network, Heart Disease Prediction, Quantum Computing, Hybrid Model, Machine Learning

I. INTRODUCTION

Cardio Vascular Diseases (CVDs), particularly heart disease, account for millions of deaths annually, representing a significant global health concern. According to the World Health Organization (WHO), nearly 17.9 million people die each year from CVDs [1], [2]. CVDs contributed to 20.5 million deaths, which is approximately one-third of all deaths across the world in 2021 alone [3]. They are the leading cause of deaths worldwide, highlighting the urgent need for improved diagnostic methods. Traditional diagnostic processes often rely on manual interpretation of patient data, which can be time-consuming and prone to human error. The rapid growth of healthcare data and advancements in computational techniques, including deep learning [4], [5] and quantum machine learning [6]–[8], offer new possibilities for more accurate and efficient medical diagnosis.

Classical machine learning methods such as logistic regression [9], Support Vector Machines (SVMs) [10], and Multi-Layer Perceptrons (MLPs) [5], [11] have been employed with varying success. Feature engineering is employed on the datasets to transform raw data into meaningful and relevant features. This is especially important in medical diagnosis using machine learning and deep learning as some features are more important than others. The lack of feature engineering can also lead to overfitting, especially on smaller datasets, which is dangerous as inaccurate predictions in a field as critical as healthcare can prove to be life-threatening. In cases where inputting patient health information into systems is done to predict heart disease, simple models that do not require extensive feature engineering work adequately. However, the necessity for faster diagnosis for early prevention and treatment measures makes these models unusable, as real-time data needs to be processed and computed within a very short amount of time.

This highlights the need for better deep learning models capable of providing fast and accurate real-time diagnosis. Recent advancements in quantum computing propose Quantum Neural Networks (QNNs) as a novel approach to handle high-dimensional data. They aim to exploit quantum properties such as superposition and entanglement [6], [7], to improve computational capability, efficiency and trainability. QNNs encode classical inputs into quantum states and employ parameterized quantum circuits as trainable models, potentially enhancing representation power and generalization.

In this study, we propose a hybrid classical-quantum heart disease prediction model integrating a QNN layer within a deep learning framework. Our model leverages the Kaggle "Heart Disease Prediction" dataset [12] and is designed to provide accurate, computationally efficient diagnoses to assist healthcare professionals in timely decision making. The remainder of the paper is structured as follows: Section II reviews related work, Section III details the methodology, Section IV presents experimentation, Section V discusses findings, and Section VI concludes with future directions.

II. RELATED WORK

Heart disease prediction has been extensively studied through classical machine learning approaches. Logistic regression, SVMs [13], decision trees, and neural networks including perceptrons and MLPs have been employed with varying success. Hybrid models such as ensemble and feature-engineered methods have also been explored [14]–[17].

Logistic regression, a supervised machine learning algorithm primarily used for binary classification (presence or absence of heart disease), finds the probability of an event occurring, which is then compared with the aim to find the higher probability value for classification. It achieved an accuracy of around 81% [15]. Decision trees are a non-parametric, supervised machine learning algorithm that creates a hierarchical, tree-like structure that simulates the human decision making process. They performed slightly lower at 79% [16].

SVMs are another supervised machine learning algorithm that computes an optimal hyperplane that separates data belonging to different classes, with the goal of guaranteeing the highest possible margin (distance) between the hyperplane and the closest data points of each class. SVM combined with ANN reached comparable results [13].

Multilayer Perceptrons (MLPs) demonstrated improved performance with accuracies near 83% [14]. These algorithms, along with neural networks including perceptrons and MLPs have been employed in heart disease prediction with varying levels of success.

However, these models either lacked robustness to nonlinear feature interactions or required heavy preprocessing. Quantum machine learning introduces novel models leveraging superposition and entanglement [6], [7]. Recent work demonstrates QNN applications to diagnosis [8] and causal machine learning in healthcare [18]. Radiology applications of ML also highlight its clinical value [19], [20].

The proposed QNN-based hybrid model differs by incorporating a variational quantum circuit layer, enabling enhanced representation of feature correlations. Our results show that QNN achieved an accuracy of 82.88% and F1-score of 0.848, outperforming some classical baselines while offering quantum-enhanced generalization.

III. Methodology

A. Dataset

The dataset has been obtained from Kaggle [12] and UCI Repository [21]. It contains 918 samples, each including demographic information (like age and sex), 9 clinical features alongside a binary heart disease label indicating presence (1) or absence (0) of heart disease. A summary of the dataset is provided in Table I.

TABLE I
Clinical Features in the Kaggle Heart Disease Dataset

Column	Description
Age	Age of the patient [years]
Sex	Sex of the patient [M: Male, F: Female]
ChestPainType	Chest pain type [TA, ATA, NAP, ASY]
RestingBP	Resting blood pressure [mm Hg]
Cholesterol	Serum cholesterol [mg/dl]
FastingBS	Fasting blood sugar [1 if ¿120 mg/dl, else 0]
RestingECG	Resting ECG results [Normal, ST, LVH]
MaxHR	Maximum heart rate achieved
ExerciseAngina	Exercise-induced angina [Y/N]
Oldpeak	ST depression (numeric)
ST Slope	Slope of ST segment [Up, Flat, Down]
HeartDisease	Output class [1: heart disease, 0: normal]

B. Data Preprocessing

Data preprocessing is an essential task to be performed before training the model to ensure reliability of the model. To ensure quality and reliability of the inputs, the dataset was subjected to a systematic preprocessing pipeline before model training.

The missing values were first handled, by imputing based on the type of attribute. For features with categorical values, the most frequent value (mode) replaced the missing values. For features with continuous values, the missing data was replaced by the mean of the remaining values in the column. This method ensured that the missing values were substituted by the most probable values, reducing the possibility of introducing outliers.

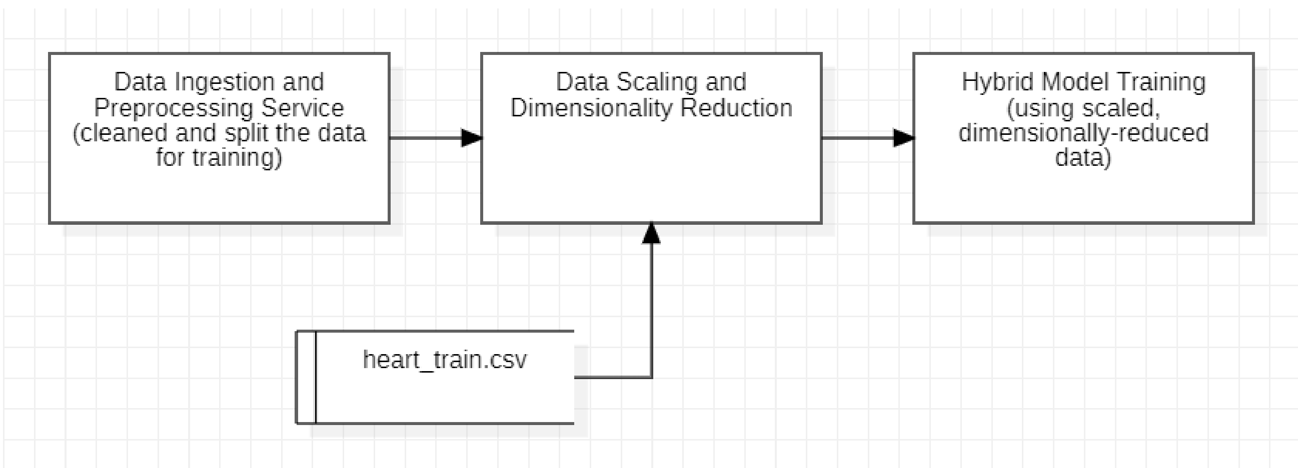

Figure. 1a. System Architecture: Training Pipeline

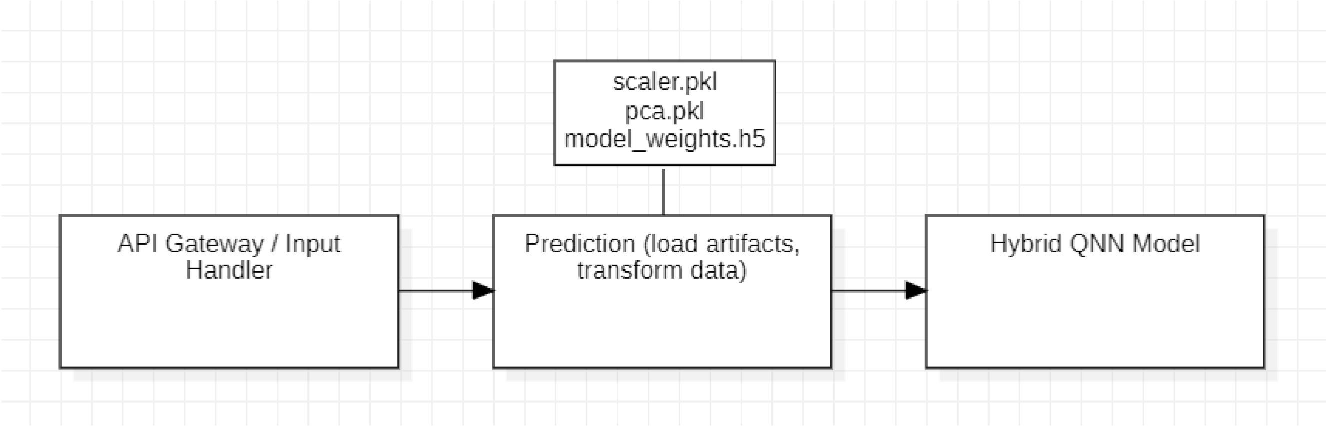

Figure. 1b. System Architecture: Inference Pipeline

Categorical features were then transformed into numerical values to enable computations to be performed on it. The *scikit-learn* package's *LabelEncoder* was employed, and each category (label) was mapped to a numerical value.

The continuous attributes were then normalized to put them on a uniform scale. Incorporating parameters like blood pressure and cholesterol directly into the model may lead to unstable gradient updates because they operate on different magnitude ranges. We used *StandardScaler* from the *scikit-learn* package, which standardizes features by transforming them to have a mean of 0 and standard deviation of 1. This method ensures that all features contribute to the training of the model proportionally. In order to apply the same transformation consistently to fresh input samples during inference, the scaler that was fitted to the training data was serialized into a pickle file.

The final step was to split the dataset into training and testing subsets. The dataset was split with 70% of the records being allotted to training the model and the remaining 30% being allotted to the testing subset. To maintain the balance of number of positive and negative cases in both subsets, the stratified sampling method was utilized. It is important to use this method as the dataset contains the records of both patients with and without heart disease, and preserving this balance prevents bias and helps the model generalize effectively.

C. *Quantum Neural Network Architecture*

The model is designed to be a hybrid classical-quantum neural network, where conventional deep learning layers were improved with a variational quantum circuit. The input consists of a vector of 11 dimensions, each dimension representing a feature of the dataset after preprocessing.

The classical part of the model begins with fully connected dense layers that use the ReLU activation function to extract higher-level representations of the input. These layers are responsible for learning non-linear relationships and patterns between the features. This provides a strong baseline for feature transformation before quantum computing is introduced.

The quantum component is implemented using *PennyLane* integrated with *TensorFlow/Keras*. The circuit is built with 8 qubits and 2 layers. The data is embedded into the quantum system by angle encoding using *PennyLane*'s *AngleEmbedding*, which maps classical inputs onto quantum states through parameterized rotations.

Within each layer, single-qubit rotations introduce variability across quantum state space, while a sequence of controlled NOT (CNOT) gates create entanglement between adjacent qubits. This entanglement allows the quantum circuit to capture correlations across several features simultaneously by exploiting quantum superposition to explore a richer representational space than would be possible with the use of classical methods.

The outputs of the quantum circuit are the expectation values (predicted average of many measurements for a particular quantum property) of the Pauli-Z operator for each qubit. This yields a 8-dimensional feature vector that is quantum-enhanced. These vectors serve as the transformed representation of the input. This quantum block is integrated as a trainable *Keras* layer and its rotation parameters are optimized along with the weights of the classical dense layers using backpropagation.

Another block of classical dense layers follow the quantum circuit for further processing. A fully connected dense layer with 16 neurons and ReLU activation function introduces more non-linear transformations by refining the quantum-enhanced features. A dropout with a rate of 0.3 is introduced to reduce the risk of overfitting. This randomly disables 30% of neurons during training, improving the robustness of the model on unseen data.

Finally, the output layer consists of a single neuron equipped with a sigmoid activation function. This layer produces an output (probability score) between 0 and 1, representing the presence or absence of heart disease based on a patient's data. Predictions are made by applying a threshold value of 0.5, i.e., the model classifies an output greater than or equal to 0.5 as presence of heart disease and less than 0.5 as absence of heart disease. The various stages in the system architecture are represented in the Figures 1a and 1b.

D. Training and Optimization

The training pipeline began with ingestion of the dataset and preprocessing it, where the data is cleaned and split into training and testing data. The features were then standardized and reduced dimensionally to improve efficiency and ensure stability in gradient updates.

The model was trained on this processed data using the binary cross-entropy loss function, suitable for binary classification tasks. The *Adam optimizer* was employed with a learning rate of 5×10^{-4} to ensure efficient and adaptive gradient updates.

To avoid overfitting, an early stopping criterion monitored validation loss and halted training when no improvement was observed over five consecutive epochs. The network was trained with a mini-batch size of 16 for up to 30 epochs, balancing computational efficiency with learning stability. Dropout layers in the dense blocks further improved generalization.

The quantum layer was simulated using the *PennyLane* and *Qiskit* frameworks on classical CPUs. During training, gradients from the quantum layer were computed using the parameter-shift rule, enabling seamless integration with back- propagation. This hybrid optimization process allowed the model to learn effective representations by combining classical gradient descent with quantum variational circuit tuning.

Trained artifacts such as the scaler and weights were saved, and used for prediction on unseen data later in the inference pipeline.

IV. Experimentation

A. Results

The output consists of a user interface depicting a form with fields to input values for each of the features in the dataset. After entering the measurements and observations of the various parameters of the patient's health and asking the model to predict, the system processes the data and produces a probability score. This probability helps the model classify the result as "At Risk" or "Not at Risk". Figure. 2a. shows the output of the prediction as "At Risk" which corresponds to a low heart health score, while Figure. 2b. depicts a classification of "Not at Risk" corresponding to a high heart health score.

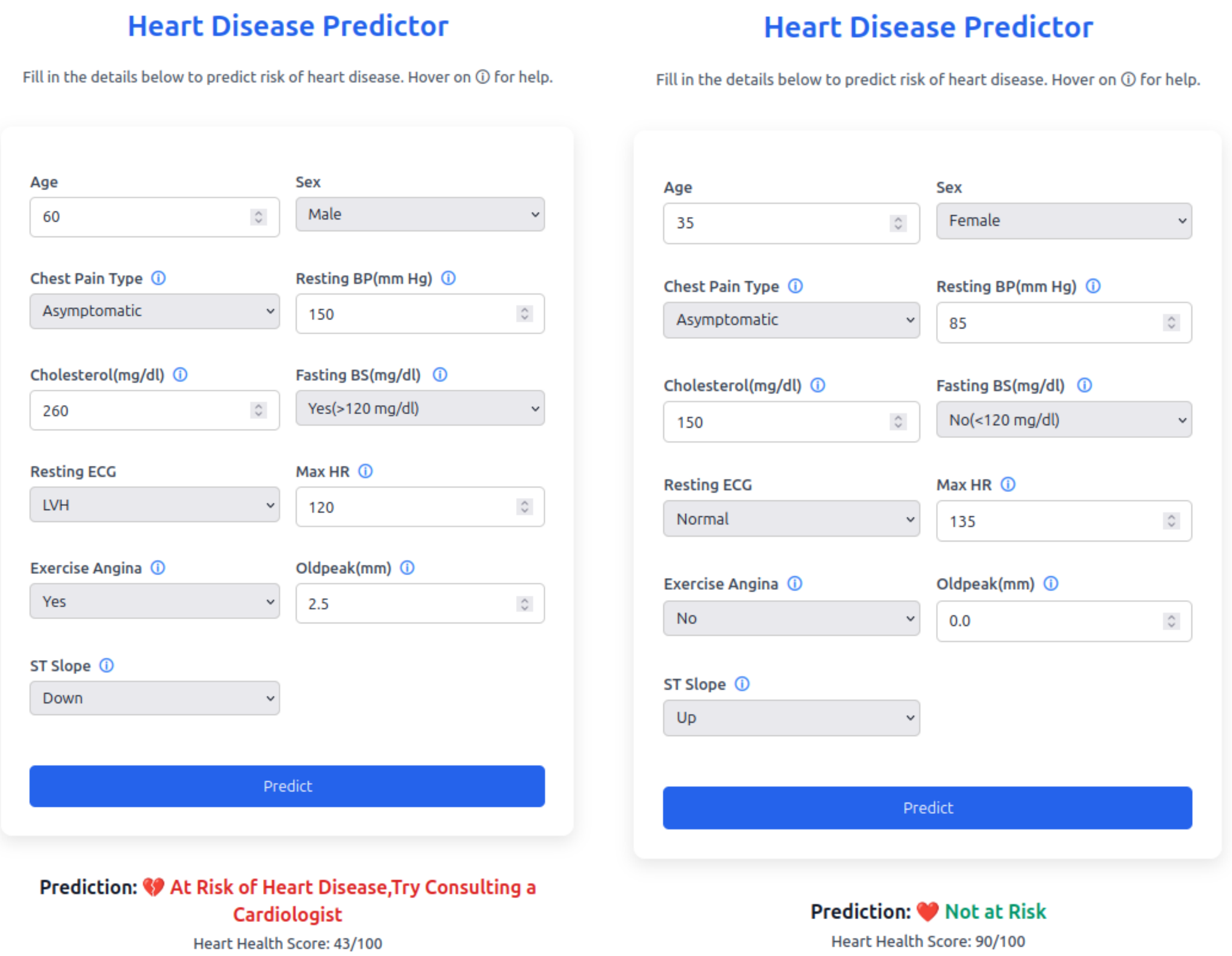

Figure. 2a. Output screen when the features indicate presence of heart disease, Figure. 2b. Output screen when the features indicate absence of heart disease

B. Performance Comparison

Comparison of the proposed hybrid quantum model with classical baselines like Logistic Regression, Decision Tree and Multilayer Perceptron is done. The evaluation is based on three key metrics: Accuracy, F1-Score, and Area Under the ROC Curve (AUC). The results are shown in Table II.

TABLE II
PERFORMANCE COMPARISON OF QNN AND CLASSICAL BASELINES

Model	Accuracy	F1-Score	AUC
Logistic Regression	0.81	0.82	0.84
Decision Tree	0.79	0.80	0.82
Multilayer Perceptron (MLP)	0.83	0.84	0.86
Quantum Hybrid (QNN)	0.8288	0.848	0.88

C. Interpretation of Metrics

Accuracy measures the proportion of correctly classified cases (both positive and negative) out of the total samples. While useful for an overall view, it may be misleading in imbalanced datasets. For instance, a model predicting all patients as "No Heart Disease" could still achieve high accuracy if negatives dominate the dataset.

Accuracy is measured using the formula:

$$Accuracy = \frac{T_P + T_n}{T_P + T_n + F_P + F_n} \quad (1)$$

where, T_p indicates true positive, T_n indicates true negative, F_p indicates false positive, F_n indicates false negative.

F1-Score is the harmonic mean of precision and recall, providing a balanced measure of the model's ability to detect both positive (heart disease) and negative (no disease) cases. Precision is the metric that measures the accuracy of positive predictions made by a model. Recall (or sensitivity) measures the model's ability to find all the positive instances.

Precision is calculated as:

$$Precision = \frac{T_P}{T_P + F_P} \quad (2)$$

where, T_p indicates true positive and F_p indicates false positive.

Recall is calculated as:

$$Recall = \frac{T_P}{T_P + F_n} \quad (3)$$

where, T_p indicates true positive and F_n indicates false negative.

F1-Score would then be:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (4)$$

The QNN achieves an F1-score of 0.848, which highlights its balanced classification capability. This is especially critical in medical applications where both false negatives (missed diagnosis) and false positives (unnecessary alarms) carry serious consequences.

AUC (Area Under the ROC Curve) evaluates the model's ability to distinguish between classes at various threshold settings. An AUC closer to 1 indicates strong discriminative power. The QNN's AUC of 0.88 suggests that it maintains superior sensitivity-specificity trade-offs compared to classical models.

D. Confusion Matrix

Table III shows the confusion matrix for the proposed QNN model, indicating its performance in correctly identifying patients with and without heart disease.

TABLE III
CONFUSION MATRIX OF QNN MODEL

	Predicted No	Predicted Yes
Actual No	39	11
Actual Yes	8	53

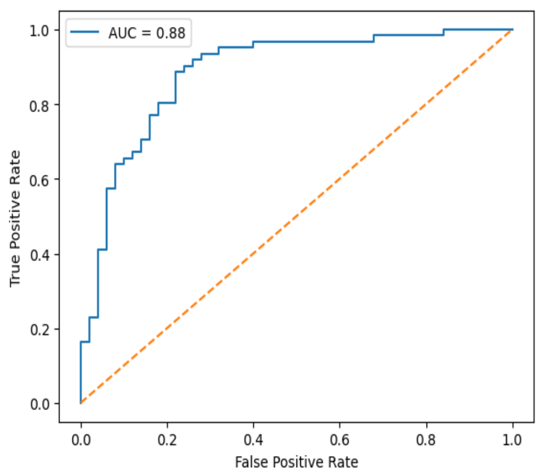

Figure. 3. ROC curves for quantum hybrid and classical models.

E. ROC Curve

The ROC (Receiver Operating Characteristic) curve is a graph that plots the true positive rate (TPR) against the false positive rate (FPR) at various classification thresholds for a binary classification model.

It visually evaluates the ability of a model to distinguish between classes by showing the trade-off between TPR and FPR. A model with good performance has a ROC curve that rises quickly to the top-left corner.

The ROC curves (Figure 3) compare the QNN against classical baselines, illustrating superior sensitivity-specificity balance.

V. Discussion

Our results demonstrate that QNN provides competitive accuracy compared to classical models while achieving stronger F1 and AUC values. The key improvement lies in the ability of quantum circuits to model complex correlations without requiring manual feature engineering. These results align with prior findings in medical AI [18], [19], showing potential of quantum-enhanced models for diagnostic support. Similar advances are seen in cardiology imaging [19] and large-scale heart disease studies [2].

However, there are limitations to leveraging the suggested model. The quantum neural network was implemented using a simulator, whose primary limitation is the exponential increase in memory and computational time required. Other concerns include execution time and costs, as evaluating quantum circuits is highly computationally expensive and takes significant amount of time. The tuning of hyperparameters also proves to be a challenge as careful selection of hyperparameters is required to achieve high results while maintaining low costs.

Further study might utilize larger datasets obtained from various sources to train QNN models more effectively for better accuracy. With the surge in research related to quantum computing, real quantum hardware can be used to implement QNN models.

VI. Conclusion

This study demonstrates that a hybrid deep learning model with QNN can effectively predict heart disease with 82.88% accuracy and an F1-score of 0.848. Such models can serve as supplementary tools for early diagnosis, enabling healthcare professionals to prioritize high-risk patients. Future work may focus on developing approaches that provide fast diagnoses based on real-time data, integrating lifestyle and genetic features for better understanding of the extent of risk of heart disease, and exploring explainable AI techniques to improve clinical trust.

Integration with wearable health-monitoring devices can improve the response time for prevention and treatment in cases of high-risk heart patients, contributing to a reduce in mortality caused by heart disease.

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